MILITARY SPECIFICATION

MULTIVITAMIN AND SODIUM FLUORIDE SOLUTION

This specification is approved for use by the Defense Personnel Support Center, Defense Logistics Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Multivitamin and Sodium Fluoride Solution (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

The National Formulary (NF).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a multivitamin and sodium fluoride solution containing in each dose the following ingredients:

Fluoride (as Sodium Fluoride)	0.5 mg
Vitamin A	1250 to 2000 International Units
Vitamin D (Ergocalciferol)	200 to 400 International Units
Vitamin E	0 to 15 International Units
Vitamin C (Ascorbic Acid)	20 to 60 mg
Vitamin B_1 (Thiamine Hydrochloride)	0.33 to 1.05 mg
Vitamin B_2 (Riboflavin as a suitable salt)	0.4 to 1.2 mg
Niacinamide	4.5 to 13.5 mg
Vitamin B_6 (Pyridoxine Hydrochloride)	0.33 to 1.05 mg
Pantothenic Acid	0 to 10 mg
Vitamin B_{12}	0 to 4.5 mcg

3.1.1 The finished solution shall contain a sufficient excess of each vitamin so as to comply with the inital assay limits (see 3.2) for each vitamin.

3.1.2 Shall be suitable for pediatric use.

3.2 Assay limits.

3.2.1 Fluoride. The finished solution shall assay to contain not less than 90.0 percent and not more than 110.0 percent of the required amount of fluoride, when determined potentiometrically using a fluoride-specification-indicating electrode.

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3.2.2 Vitamin A. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of vitamin A, when determined using the general USP method for vitamin A assay.

3.2.3 Vitamin D. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of vitamin D, when determined using the general USP method for vitamin D assay.

3.2.4 Vitamin E. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of vitamin E, when determined using the general USP method for alpha tocopheral assay.

3.2.5 Vitamin C. The finished solution shall assay to contain not less than 100.0 percent of the required amount of vitamin C when determined using the USP method for Ascorbic Aicd Injection, making any necessary sample size or dilution changes, as required.

3.2.6 Thiamine hydrochloride. The finished solution shall assay to contain not less 100.0 percent of the labeled amount of thiamine hydrochloride, when determined using the general USP method for thiamine assay.

3.2.7 Riboflavin. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of riboflavin, when determined using the general USP method for riboflavin assay.

3.2.8 Niacinamide. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of niacinamide, when determined using the general NF method for niacin and niacinamide.

3.2.9 Pyridoxine hydrochloride. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of pyridoxine hydrochloride, when determined as specified in the USP monograph for Decavitamin Capsules. Sample size and dilution changes to accommodate the assay of the solution shall be made, as required.

3.2.10 Pantothenic acid. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of pantothenic acid when determined by a suitable, accurate, reproducible method.

3.2.11 Vitamin B_{12}. The finished solution shall assay to contain not less than 100.0 percent of the labeled amount of vitamin B_{12} when determined by a suitable, accurate

3.3 Alternate methods. Alternate assay methods and test procedures to those designated in 3.2 are permitted.

3.4 Ingredients. All ingredients used in the manufacture of the finished solution shall be in accordance with the tests, standards, and requirements of the USP or NF, including any supplements or revisions thereto; or, if not included in either of these compendia, the ingredients shall be of the highest pharmaceutical grade and shall be suitable for use in the solution.

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3.5 Expiration dating period. The expiration dating period shall be not less than 18 months. Not less than 100.0 percent of the labeled amount of each vitamin shall remain during the labeled expiration dating period of the solution.

3.5.1 Delivery. Not more than 3 months of the expiration dating period shall have elapsed at the time of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of the inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of solution produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

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4.4 Examination. The finished solution shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects.

TABLE II. Classification of defects.

Categories	Defects 1/
Major	
101	Solution not free from foreign matter.
102	Solution has crystal formation.
103	Solution not homogeneous.
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks or tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests. Tests shall be performed to determine compliance with all requirements of this document and all referenced documents.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186, and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.2.1 Unit of issue. One bottle consisting of one bottle and dropper, as specified, constitutes one unit of issue.

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5.2.2 Level A.

5.2.2.1 Immediate container. The immediate container shall be a commercially available bottle with screw cap dropper, calibrated to provide recommended dose, considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other. As an option, the dropper may be supplied separately in the unit package.

5.2.2.2 Unit package. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate package. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The packaged commodity shall be packed in shipping containers that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the expiration date.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.
(b) the National Stock Number.

NOTES: (1) The National Stock Number need not be preceded by any prefix.
(2) Strip labeling may be used.
(3) Label adhesion test shall not apply.
(4) If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

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5.4.2 Intermediate package. Intermediate package shall be marked as specified in 5.5.3 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

5.4.3 Exterior container. Exterior container shall be marked as specified in 5.5.4 of PPP-C-186. Type I shelf life marking as specified in MIL-STD-129 shall be shown. Marking shall include the legend:

"STORE AT CONTROLLED ROOM TEMPERATURE 15° - 30°C (59° - 86°F)".

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) National Stock Number.
(c) Selection of applicable levels of packaging and packing (see 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-01-031-3195	MULTIVITAMIN AND SODIUM FLUORIDE SOLUTION, 50 Doses

Preparing activity:
DLA - DM

Project No. 6505-1980

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DEFENSE PERSONNEL SUPPORT CENTER
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

DEFENSE PERSONNEL SUPPORT CENTER
DIRECTORATE OF MEDICAL MATERIEL, CODE-ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA 19101

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-M-37830(DLA-DM) MULTIVITAMIN AND SODIUM FLUORIDE SOLUTION

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

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